



Week 1 Lecture 4

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Software Architectures

What is a design pattern?

"A general, re-usable solution to a commonly occuring problem within a given context in software design"

Source: Wikipedia

- Experienced designers observe "patterns" in code
- Re-using these patterns can make design and development faster
- Guide the design and thought process

M-V-C paradigm

- **Model:**
 - Core data to be stored for the application
 - Databases: indexing for easy searching, manipulation, etc;
- **View:**
 - User-facing side of the application
 - Interfaces for finding information, manipulating
- **Controller:**
 - "Business Logic" - how to manipulate the data

Origins in the smalltalk language, 1979

The user uses the **controller** → to manipulate the **model** → that updates the **view** → that the user sees;

Other design patterns

- Model-View-Adapter
- Model-View-Presenter
- Model-View-ViewModel
- Hierarchical MVC
- Presentation-Abstraction-Control

Each one of them has its own uses, but the fundamentals are very similar

Focus of this course

- Platform:
 - Web-based
- Architecture:
 - Client-server
- Software architecture
 - Model-View-Controller